

SOLUTION QUIZ-01 BSE-6B

QUESTION-01

1. B
2. C
3. A
4. D
5. A

QUESTION-02

Transmission Delay = $L/R = 5000 / 1000000 = 0.005\text{s} = 5\text{ms}$

Packet 1

1. **Host A sends Packet 1:** Ends at 5 ms.
2. **Propagates to Switch:** Arrives at switch at $5\text{ ms} + 0.02\text{ ms} = 5.02\text{ ms}$.
3. **Switch Processing:** $5.02\text{ ms} + 0.035\text{ ms} = 5.055\text{ ms}$.
4. **Switch forwards Packet 1:** Starts at 5.055 ms. Takes 5 ms to transmit. Ends at 10.055 ms.
5. **Propagates to Host B:** Arrives at destination at $10.055\text{ ms} + 0.02\text{ ms} = 10.075\text{ ms}$.

Packet 2 (Pipelining)

Host A sends Packet 2 immediately after Packet 1.

1. **Host A sends Packet 2:** Starts at 5 ms, ends at 10 ms.
2. **Propagates to Switch:** Arrives at switch at $10\text{ ms} + 0.02\text{ ms} = 10.02\text{ ms}$.
3. **Switch Processing:** $10.02\text{ ms} + 0.035\text{ ms} = 10.055\text{ ms}$.
4. **Switch forwards Packet 2:**
 - The switch finished sending Packet 1 at exactly 10.055 ms (from Step 2).
 - The switch is free, and Packet 2 is processed and ready at exactly 10.055 ms.
 - Transmission starts at 10.055 ms and takes 5 ms.
 - Transmission ends at 15.055 ms.
5. **Propagates to Host B:** $15.055\text{ ms} + 0.02\text{ ms} = 15.075\text{ ms}$.

Final Answer: 15.075 ms.